STACKS, DEFORMATION THEORY: LEMMA-MATCH-THICKENINGS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4873-4889 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 4891-4931, the complete author proof (39 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 16 of queue_defos.json (pb=1853, stmt 4873-4889, proof 4891-4931).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 08M6: sources/stacks-defos/pinned-original-source.tex, source lines 2669–2687. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D06: sources/stacks-defos/pinned-original-source.tex, source lines 4931–4931. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07V6: sources/stacks-defos/pinned-original-source.tex, source lines 4931–4931. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZS: sources/stacks-defos/pinned-original-source.tex, source lines 4931–4931. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZJ: sources/stacks-defos/pinned-original-source.tex, source lines 4931–4931. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03M9: sources/stacks-defos/pinned-original-source.tex, source lines 4931–4931. Cited by the retained statement or proof but not part of

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the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03G8: sources/stacks-defos/pinned-original-source.tex, source lines 4931–4931. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $i : Z \to Z'$ be a morphism of algebraic spaces over S. The following are equivalent*

(1) *i is a thickening of algebraic spaces as defined in More on Morphisms of Spaces, Section 05ZJ, and*
(2) *the associated morphism $i_{small} : (Sh(Z_{étale}), \mathcal{O}_Z) \to (Sh(Z'_{étale}), \mathcal{O}_{Z'})$ of ringed topoi (Properties of Spaces, Lemma 03G8) is a thickening in the sense of Section 08M6.*

Proof. We stress that this is not a triviality.

Assume (1). By More on Morphisms of Spaces, Lemma 05ZS the morphism i induces an equivalence of small étale sites and in particular of topoi. Of course $i^\sharp$ is surjective with locally nilpotent kernel by definition of thickenings.

Assume (2). (This direction is less important and more of a curiosity.) For any étale morphism $Y' \to Z'$ we see that $Y = Z \times_{Z'} Y'$ has the same étale topos as Y'. In particular, Y' is quasi-compact if and only if Y is quasi-compact because being quasi-compact is a topos theoretic notion (Sites, Lemma 0D06). Having said this we see that Y' is quasi-compact and quasi-separated if and only if Y is quasi-compact and quasi-separated (because you can characterize Y' being quasi-separated by saying that for all Y'_1, Y'_2 quasi-compact algebraic spaces étale over Y' we have that $Y'_1 \times_{Y'} Y'_2$ is quasi-compact). Take Y' affine. Then the algebraic space Y is quasi-compact and quasi-separated. For any quasi-coherent $\mathcal{O}_Y$-module $\mathcal{F}$ we have $H^q(Y, \mathcal{F}) = H^q(Y', (Y \to Y')_*\mathcal{F})$ because the étale topoi are the same. Then $H^q(Y', (Y \to Y')_*\mathcal{F}) = 0$ because the pushforward is quasi-coherent (Morphisms of Spaces, Lemma 03M9) and Y is affine. It follows that Y' is affine by Cohomology of Spaces, Proposition 07V6 (there surely is a proof of this direction of the lemma avoiding this reference). Hence i is an affine morphism. In the affine case it follows easily from the conditions in Section 08M6 that i is a thickening of algebraic spaces. □